## ADVANCED PROBLEMS FOR LECTURE 22

In this final lecture for the course, we look back at the material we learned in the course. So we will use this advanced problem sheet to instead look ahead at what's to come. For the past several lectures, the advanced problems have been an attempt to open a few doors to a new world of mathematics. You can walk through any of them to explore whatever seems appealing. There are lots of profound patterns to investigate and be amazed by! Here are a few more ideas that will come up in your studies soon.

(1) **Sequences and Series** We only really studied limits of functions, but there is a corresponding notion of a limit of a *sequence*. We got a bit of a taste of sequences when we were dealing with a sequence of Riemann sums which converge to an area. Sequences are a bit less "visual", but a little more straight forward to understand.
   (a) Let $a = (a_1, a_2, a_3, \ldots)$ be a sequence. We sometimes call the definition of limit of this sequence the $\epsilon$-$N$ limit definition. See if you can come up with what this definition should be, by extrapolating the idea from our $\epsilon$-$\delta$ definition for functional limits.
   (b) A series $A = a_1 + a_2 + a_3 + \cdots$ is said to *converge* if its sequence of partial sums $b = (b_1, b_2, b_3, \ldots)$ where $b_k = a_1 + a_2 + \cdots + a_k$ has a finite limit. Show using your definition from (a) that the series
   $$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$$
   converges to 2.
   (c) A famous series is the *harmonic series*, defined as
   $$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots$$
   Does this series converge?

(2) **Taylor Approximations** We saw in a previous advanced problem set that we don't have to limit ourselves to linear approximations if we know our function can be differentiated more than once. In general, if $f$ is an $n$-times differentiable function, the "Taylor approximation of order $n$ to $f$ at $a$" is the polynomial function
$$T_{n,a}(x) = \sum_{k=0}^{n} f^{(k)}(a) \frac{(x-a)^k}{k!}.$$
   (a) Show that the Taylor approximation of degree 1 is simply the tangent line (or the usual linear approximation.)
   (b) See if you can derive the Taylor polynomial approximation by noting that if $f$ is differentiable $n$ times, then $f^{(n-1)}$ is differentiable (at least) once, so it can be approximated by a tangent line, then integrating.
   (c) In the case where our function is infinitely differentiable, we can extend our Taylor polynomials to a *Taylor series*:
   $$T_a(x) = \sum_{k=0}^{\infty} f^{(k)}(a) \frac{(x-a)^k}{k!}$$
   Is it true that if $f$ is infinitely differentiable, then
   $$f(x) = T_a(x)$$

for all $x$ near $a$? (This is more of a rhetorical question at the moment. It is not at all easy to answer, especially since we don't have an appropriate notion of convergence.)

(3) **Multivariable Calculus** We have been dealing almost entirely with single variable functions. But more complex phenomena require more complicated models, which very often means more variables. We can still do calculus in this new multi-dimensional setting, but we should be very careful about what we mean.

Lets define a $n$-variable function to be a function whose domain is (some subset of) $\mathbb{R}^n$ and whose codomain is $\mathbb{R}$ (could also make it $\mathbb{R}^m$ instead). That is if $\overline{x} = (x_1, x_2, \ldots, x_n)$ is in the domain of $f$, then we write

$$f(\overline{x}) = f(x_1, x_2, \ldots, x_n)$$

The *level curves* of $f$ are the sets $C_z := \{\overline{x} : f(\overline{x}) = z\}$, indexed by real numbers $z$ in the range of $f$.

(a) Let's warm up with a simple example. Consider the function $f(x, y) = x^2 + y^2$. Show that the level curves of $f$ either have one point, or are circles.

(b) An issue arises immediately if we try to do calculus with these functions however. What goes wrong if we naively try to invoke the definition of derivative for these functions?

(c) Still, the notion of a derivative exists, however it is no longer of the same "type" as our original function, like it was in the single variable case. And it can often be reconstructed with the notion of *partial derivatives*. The idea is simply to take the derivative of our function with respect to one of our variables, while treating the others as a constant. The notation we use for this is $\frac{\partial}{\partial x}$ (for a partial derivative with respect to $x$) and $\frac{\partial}{\partial y}$ (for a partial derivative with respect to $y$). Calculate the partial derivatives of our function $f$ given in part (a). Show also that

$$\frac{\partial}{\partial x}\frac{\partial}{\partial y} f(x, y) = \frac{\partial}{\partial y}\frac{\partial}{\partial x} f(x, y).$$

(4) **Higher Dimensional Topology** We saw in a previous advanced problem sheet, the definition of a topology being that of a collection of open sets. We proceeded to investigate a bit of topology on $\mathbb{R}$. But we can of course extend this idea to $\mathbb{R}^n$ as well. The open sets here are a bit more visually interesting as well, but because of the added dimensions, a bit harder to wrap our heads around too.

Pick some $x \in \mathbb{R}^n$ (so $x$ is a vector now) and $r > 0$. Define the *open ball of radius $r$ centered at $x$* to be

$$B_r(x) := \{y \in \mathbb{R}^n : \|y - x\| < r\}$$

where $\|v\|$ is the Euclidean norm: $\|v\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$.

Define a set $S \subseteq \mathbb{R}$ to be *open* if whenever $x \in S$, then there is some $r > 0$ such that $B_r(x) \subseteq S$.

(a) Show that open balls are open sets. (This is less trivial than it looks).

(b) We say a set $C$ is *closed* if $C = \mathbb{R}^n \setminus S$ for some open set $S$. Show that a set $C$ is closed if and only if for any sequence $(\overline{x_n})_{n\in\mathbb{R}} \subseteq C$ that converges to some point $\overline{x}$, then $\overline{x} \in C$.

(c) Try to come up with a notion of continuity for functions $f : \mathbb{R}^n \to \mathbb{R}^m$. (Hint: you can essentially just use the $\epsilon$-$\delta$ definition, but intead of absolute values, use the Euclidean norm.)

(5) **Metric Spaces** If you want an even bigger picture of what's happening, studying metric spaces might be for you. The simple ideas of metric space theory explain why we see $\epsilon$'s and $\delta$'s show up everywhere in calculus, and provides a nice stepping stone to some more abstract topology.

A metric space is just a set that comes with a "ruler", or a *distance function*. Formally, a *metric space* is a pair $(X, d)$ where $X$ is some non-empty set, $d : X \times X \to \mathbb{R}^{\geq 0}$, such that for all $x, y, z \in X$,

(i) $d(x, y) = 0$ if and only if $x = y$,
(ii) $d(x, y) = d(y, x)$, and
(iii) $d(x, z) \leq d(x, y) + d(y, z)$.

(a) Check that $(\mathbb{R}, d)$ is a metric space, where $d(x, y) = |x - y|$.
(b) Check that $(\mathbb{R}^n, d)$ is a metric space, where $d(x, y) = \|x - y\|$.
(c) Come up with a notion of "ball of radius r" that works in a general metric space by looking at the definition we used in $\mathbb{R}^n$ (see above).
(d) Come up with a notion of *limit* for a sequence in $(X, d)$ by abstracting your idea in the first question. (Hint: Instead of absolute values, use the distance function $d$.)

These are just a small taste of what's to come, if you're interested! The world of maths is much wider and much more profound than perhaps you were lead to believe in high school. Happy hunting!